ETHANE CRACKING FURNACE · RUN LENGTH

HOW MANY DAYS DOES YOUR FURNACE REALLY HAVE LEFT?

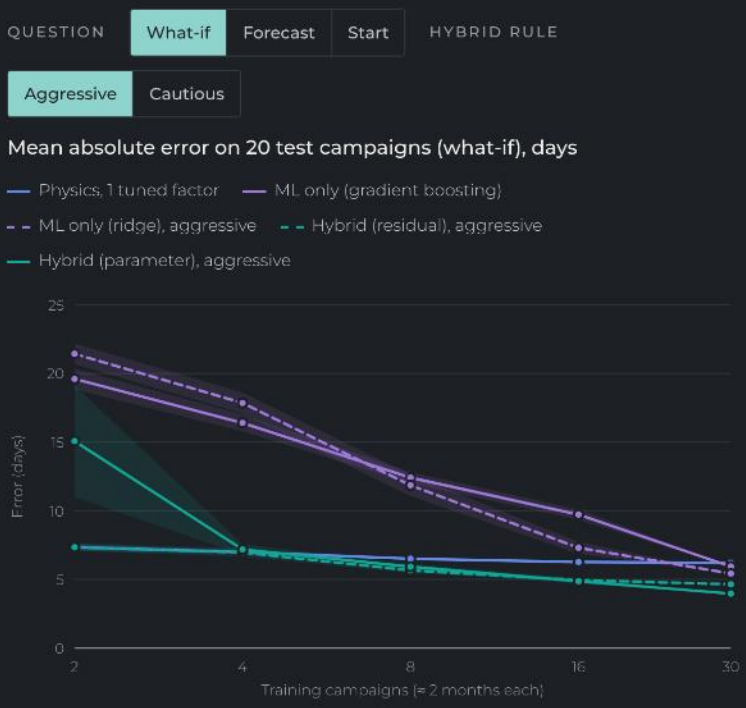
And what does guessing wrong cost? I tested physics, ML and a hybrid on a furnace where the right answer is known.



Live € calculator from the project page. Without a spare furnace, early "safe" decokes cost €130–420k per furnace-year with any model (€25–77k with a spare). The hybrid's extra saving: –€12k to +€123k, depending on set-up and margin.

WITH LITTLE DATA, PHYSICS WINS

With 2 to 8 campaigns, ML alone was 1.3 to 3 times worse than a physics model with one tuned factor. Even with 30 campaigns, about five years of one furnace, it had not clearly caught up.



Each training size up to 16 is drawn 20 times at random; bands show ± 1 standard error. At 30 there is only one possible training set, so no band. Penalties are chosen by leave-one-out on training data only. The rule toggle switches the hybrids and ML ridge; boosting has no penalty to choose.

HONEST LIMITS

WHAT THIS DOES NOT SHOW

- One synthetic plant that shares the physics model's chemistry; real plants differ more. No plant, client or employer data.
- On fresh data the hybrid's gain was 0.3–0.6 days and not yet proven. It looked bigger before I checked.
- In pure forecasting no model was significantly better than physics.
- Margin: US Gulf ethane basis. Cost of one decoke: an assumption.

How do you decide when to decoke today?

Interactive page, notebook and code: link in the comments.